

Movie Industry Analytics: Revenue Drivers and Audience Reception

– A Statistical deep dive into the Movie Industry in Python & Tableau

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1.0 Project Objective

The objective of this project is to apply rigorous statistical and exploratory data analysis techniques on a real-world movie dataset to uncover meaningful patterns and relationships.

Specific Goals

- Identify the key drivers of movie ratings, popularity, and commercial success.
- Perform hypothesis testing to validate or reject assumptions about the movie industry.
- Build regression models to understand and predict revenue based on available features.
- Design an interactive Tableau dashboard that communicates insights clearly to a non-technical audience.
- Derive actionable recommendations grounded in statistical evidence.

2.0 Dataset Description

The project uses The Movies Dataset (rounakbanik, Kaggle) originally sourced from The Movie Database (TMDb) and GroupLens. The datasets span movies released between 1874 and 2020 across 20 primary genres.

The Dataset has five csv files from which the following four have been utilized in this project –

1. **movies_metadata.csv** - (45466 rows, 24 columns) containing features such as budget, revenue, and popularity along with JSON columns for production_companies and genres features.
2. **ratings.csv** - (26M rows, 4 columns) containing features like movieId, userId, movie rating.
3. **credits.csv** - (45504 rows, 3 columns) containing JSON columns crew and cast along with tmdbId.
4. **links.csv** - (45843 rows, 3 columns) containing columns movieId, tmdbId, imdbId establishing the links between movieId and tmdbId used to merge the datasets.

Data cleaning was performed on these 4 datasets to create 2 clean datasets detailed below-

Dataset	Records	Key Columns	Primary use
cleaned_dataset.csv	42,329 rows	title, release_year, primary_genre, avg_rating, review_count, runtime, popularity, director, primary_company	EDA, hypothesis testing, rating & genre analysis
financial_dataset.csv	5,365 rows	budget, revenue + all above (zeros excluded)	Regression, ROI analysis

Key Columns Glossary

avg_rating — Aggregate user rating (0.5–5.0 scale) derived from 26M+ user reviews

review_count — Number of individual user ratings submitted for the movie

popularity — TMDb proprietary popularity score based on views, votes, and watchlist adds

primary_genre — Single dominant genre assigned to the movie (20 unique genres)

all_genres — Pipe-separated full genre list for multi-genre analysis

primary_company — Lead production studio associated with the movie

budget / revenue — Production budget and worldwide box office revenue (USD)

director — Primary director credited for the movie

3.0 Data Cleaning

- Loaded all four source files using pandas' Python parsing engine to handle malformed rows caused by embedded JSON-string columns in `movies_metadata.csv` and `credits.csv`.
- **Aggregated 26 million** individual user ratings into a single row per movie, generating `avg_rating` (mean rating) and `review_count` (total votes) as analysis-ready features.
- Parsed nested JSON-style columns (`genres`, `production_companies`, and `crew`) using `ast.literal_eval` to extract structured features including `primary_genre`, `all_genres`, `primary_company`, and `director`, then removed the original raw JSON columns.
- Retained only **analysis-relevant columns** and **removed redundant identifiers** and metadata fields not required for downstream analysis.
- **Corrected datatypes** across key variables: `id`, `budget`, and `popularity` were converted to numeric formats (with invalid entries coerced and removed), while `release_date` was converted to **datetime format** and used to derive `release_year`.

- Handled **missing values** by dropping records with null values in critical analytical fields (title, release_date, primary_genre, popularity, revenue) and **imputing missing runtime** values using the **column median** to preserve valid records.
- Merged the ratings, links, metadata, and credits datasets sequentially using the links table as a bridge between MovieLens movieId and TMDb tmdbId, producing a consolidated movie-level dataset.
- **Identified and removed duplicate movie records** introduced during merging by retaining a single representative row per movieId, keeping the highest-popularity entry where minor metadata inconsistencies existed.
- Exported two analysis-ready datasets:
 - `cleaned_dataset.csv` containing 42,329 movies for EDA and hypothesis testing
 - `financial_dataset.csv` containing 5,365 movies with non-zero budget and revenue values for financial analysis and regression modeling, where zero values were treated as missing financial information rather than true zeroes.

4.0 Feature Engineering

- Created **budget_tier** categorical segments (**Low, Mid, High, Blockbuster**) using quartile-based budget thresholds to analyze financial performance across production scales.
- Engineered **studio_tier** and **studio_label** features to classify production companies as either **Major Studios** or **Independent Studios** based on production volume within the financial dataset. Studios producing at least five films were categorized as major studios.
- Created **release_season** categories (Summer, Year-End, Other) from release dates to analyze seasonal effects on movie revenue and profitability.
- Calculated **ROI (Return on Investment)** using revenue and budget data to measure financial profitability at the movie level.

$$ROI = (Revenue - Budget) / Budget$$

- **Winsorized** extreme **ROI outliers** at the 1st and 99th percentiles to create **roi_capped**, reducing the influence of highly skewed profitability values in statistical analysis.
- **Applied** log transformations to **highly skewed financial variables**, creating **log_budget** and **log_revenue** for improved statistical modeling and regression stability.

- Implemented a **weighted movie rating** metric to **reduce volatility** from low-review films by incorporating both average rating and review volume into a Bayesian-style weighted score.

$$WR = (v/(v+m)) * R + (m/(v+m)) * C$$

Where,

R = movie average rating

v = review count

C = overall average rating across dataset

m = minimum review threshold

5.0 Business Question

This project investigates the following business question within the movie industry:

What factors influence a movie's financial performance and audience reception?

The subsequent Exploratory Data Analysis (EDA), statistical hypothesis testing, and regression analysis aim to identify and validate the relationships between movie attributes — such as budget, studio scale, release timing and genre — and measures of movie success including revenue, ROI, and weighted audience ratings.

6.0 Exploratory Data Analysis

6.1 Genre Performance: Revenue and Audience Reception

This section explores how **movie genres relate to both financial success and audience reception**. Rather than focusing **only on the frequency of genres** within the dataset, the analysis evaluates genre performance using two key business metrics: **median box office revenue** and **average weighted audience rating**.

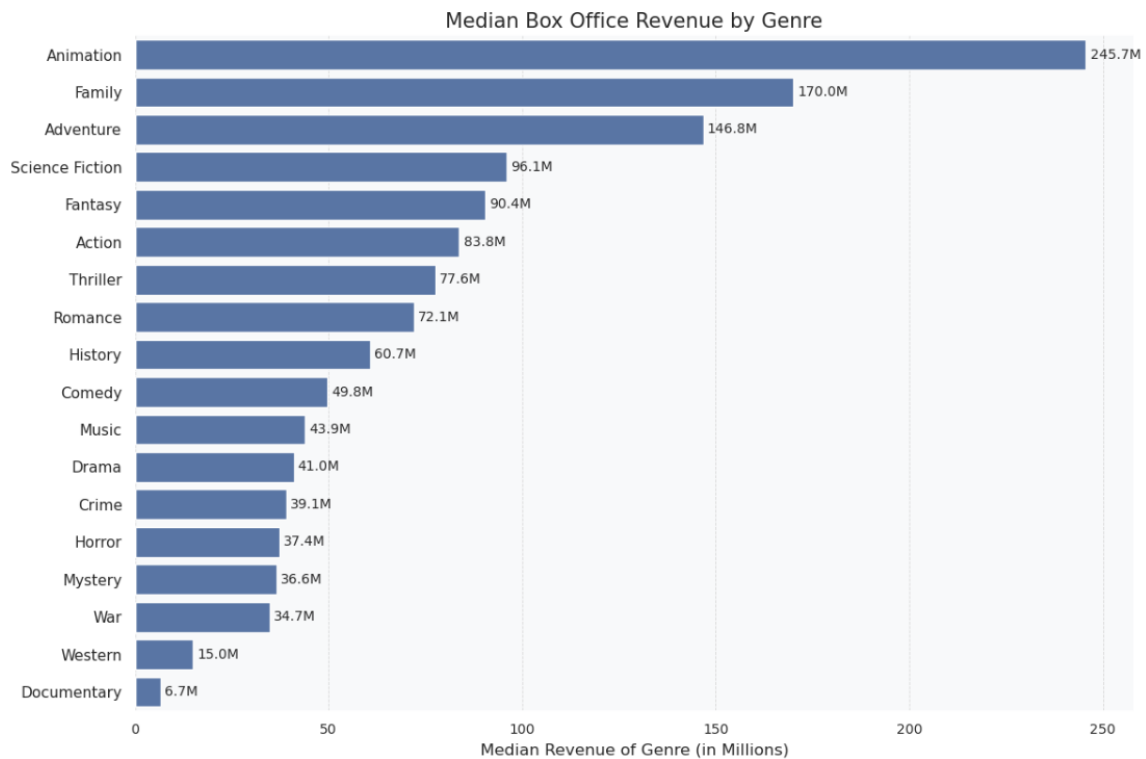
To improve the reliability of audience-based insights, only movies with a **minimum review count of 500** were included in the analysis. This filtering helps **reduce the influence of low-engagement movies** with potentially unstable ratings.

The following visualizations compare genres based on:

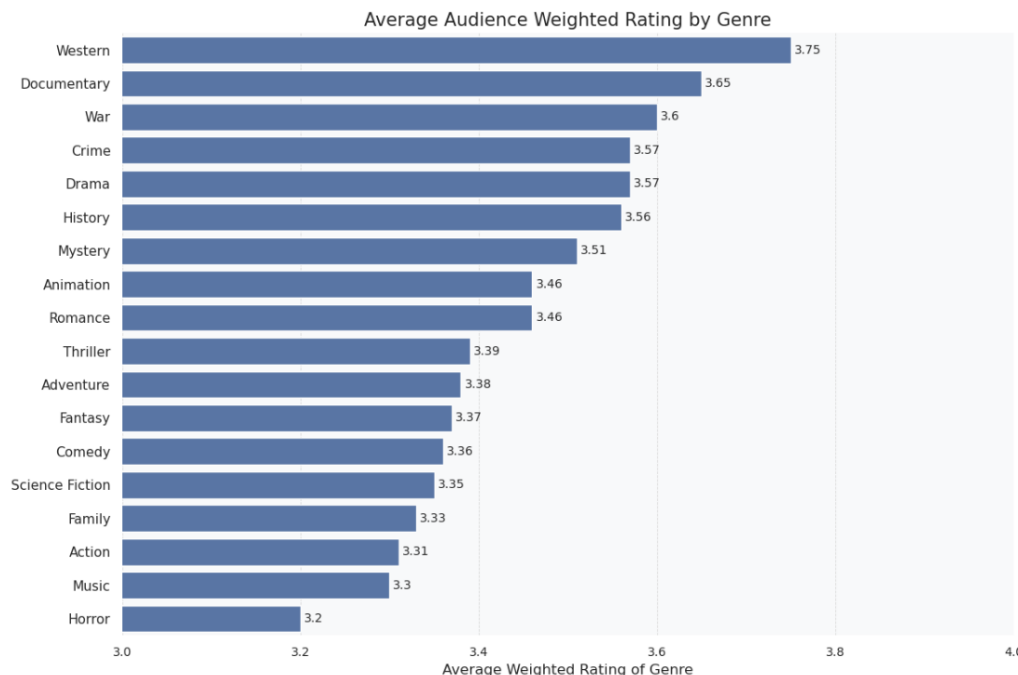
- Median Box Office Revenue (in millions)
- Average Weighted Audience Rating

These analyses provide an initial understanding of which **genres tend to perform strongly commercially**, and which **genres receive higher audience appreciation**.

Median Box Office Revenue by Genre



Average Audience Weighted Rating by Genre



Important Note Regarding Genre-Level Analysis

Some genres such as *War*, *History*, and *Western* contain **relatively small sample sizes** compared to **larger genres** such as *Drama*, *Comedy*, and *Action*. As a result, the average ratings and revenue metrics for these smaller genres may **be less stable** and should be interpreted with caution.

Insights from Genre Performance Analysis

The visualizations reveal notable differences between audience reception and commercial performance across movie genres.

Genres such as *Western*, *Documentary*, and *War* receive some of the highest average weighted audience ratings, with weighted ratings above **3.6**. However, these genres generate relatively low median box office revenues compared to commercially dominant genres. This suggests that **critically appreciated movies are not always associated with strong financial success**.

In contrast, *Animation*, *Family*, and *Adventure* emerge as the highest-performing genres in terms of median box office revenue, with median revenues ranging from approximately **\$147 million to \$246 million**. Despite their strong commercial performance, their audience weighted ratings remain comparatively moderate, **indicating broad market appeal and consistent financial success rather than exceptionally high audience acclaim**.

Genres such as *Romance* and *Thriller* demonstrate balanced performance across both financial and audience metrics, combining relatively strong median revenues with above-average audience ratings. These genres appear to **achieve both commercial viability and positive audience reception**.

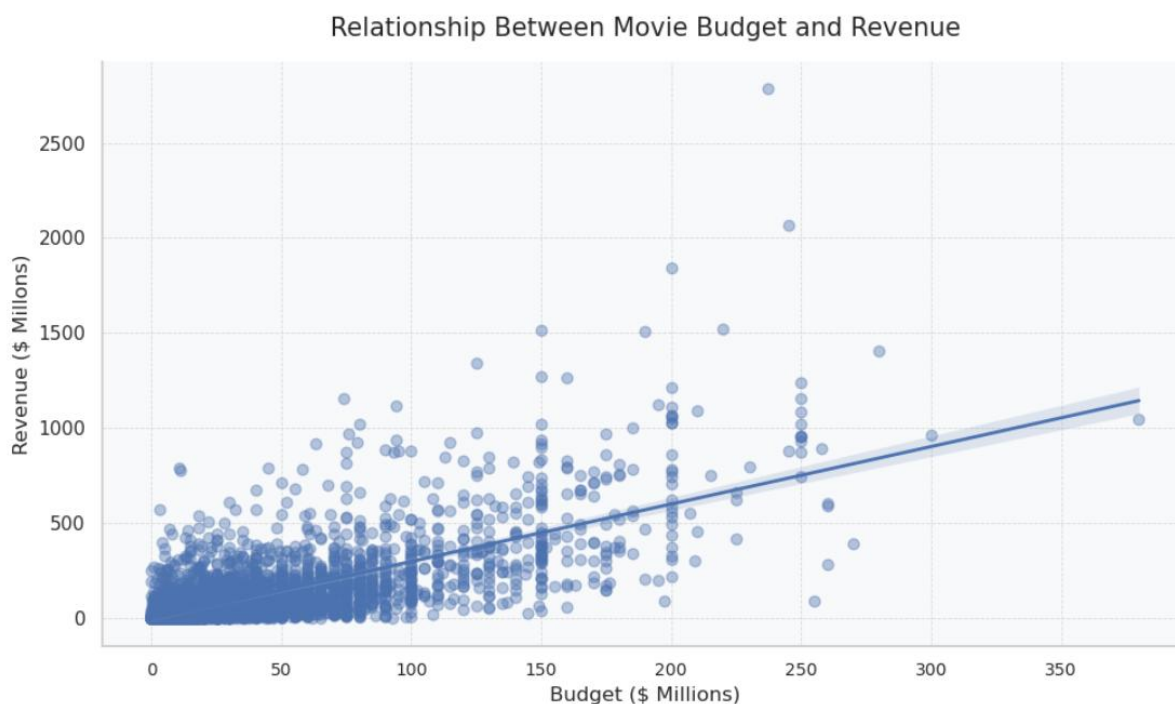
Overall, the analysis suggests that **financial success and audience appreciation do not always align across genres**, highlighting the importance of evaluating both dimensions independently when assessing movie performance.

6.2 Relationship between Movie Budget and Box office Revenue

This section explores the **relationship** between **budget** and **box office revenue** of movies to investigate whether **higher budget movies** tend to be **more successful financially**.

The following scatterplot visually depicts this relationship:

Scatterplot of Budget vs Revenue



Insights from Scatterplot Analysis

This analysis examines the relationship between movie production budgets and box office revenue to determine whether higher-budget movies tend to achieve stronger financial performance.

The scatterplot shows a clear **positive relationship** between **budget** and **revenue**, with a **correlation coefficient** of **0.73**, indicating a **moderately strong linear association between the two variables**.

While most movies follow an approximate upward trend, **several outliers are visible**, including **high-budget films with relatively low revenue** and blockbuster movies that **generated exceptionally high revenue compared to their budgets**.

Overall, the observed linear relationship suggests that **budget is an important predictor of box office revenue** and supports the **use of these variables in subsequent regression analysis**.

6.3 Top Directors and Studios by Revenue and Audience Reception

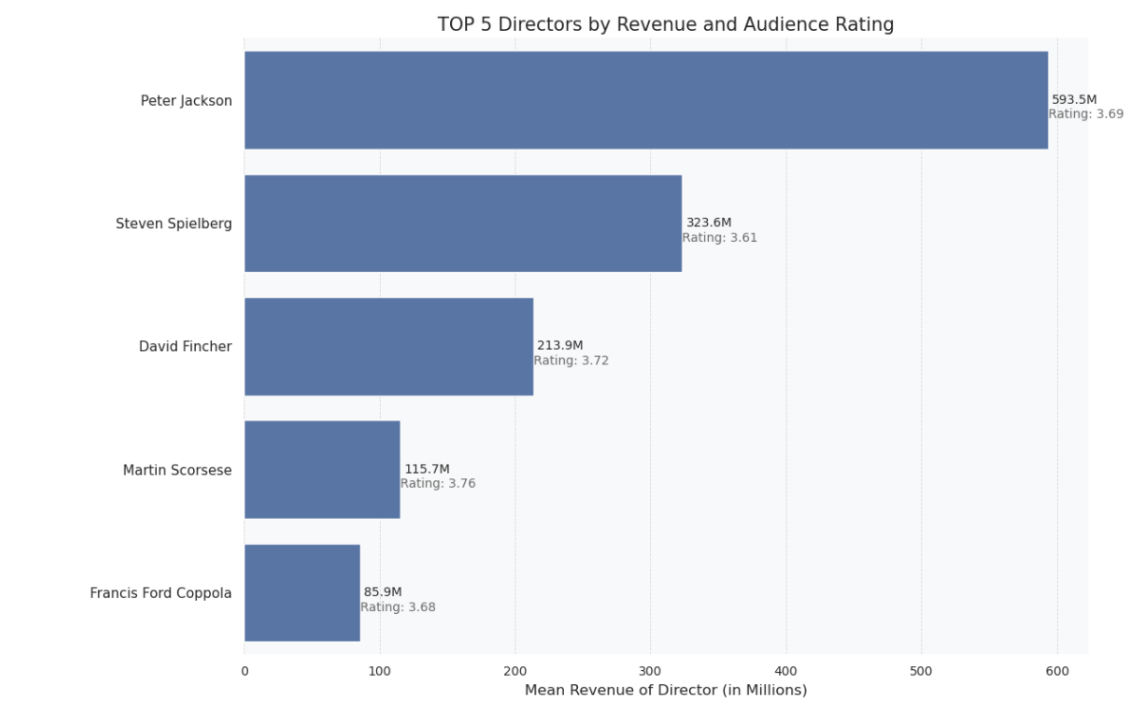
The **reputation of directors** and the **commercial strength of production studios** can both play an **important role** in influencing a movie's financial performance and audience reception. This section analyzes the **top-performing directors** and **studios** based on **average box office revenue** and **weighted audience ratings**.

To identify **consistently successful directors**, the analysis was restricted to movies with a **minimum review count of 500 to improve rating reliability**. Additionally, only directors with **at least 10 movies** in the filtered dataset were considered. Among these directors, those belonging to the **top quartile of average weighted ratings** were selected and ranked by average box office revenue.

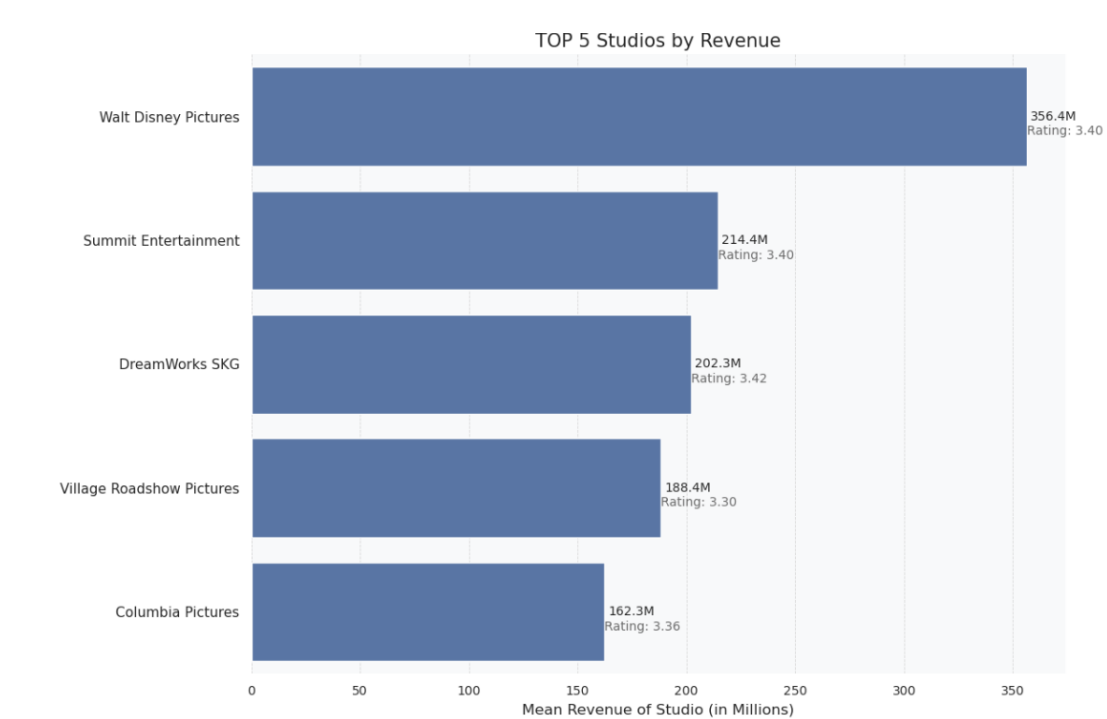
Similarly, **the analysis identifies the top-performing production studios** based on average box office revenue. To ensure **statistical stability** and avoid **small-sample distortions**, only studios with **at least 25 movies** in the filtered dataset were **included in the comparison**.

The following bar charts present the top 5 directors and top 5 studios based on these criteria.

TOP 5 Directors by Revenue and Audience Rating



TOP 5 Studios by Revenue



Key Insights from Director and Studio Analysis

Peter Jackson dominates on revenue; Martin Scorsese leads on quality. Jackson's mean revenue of \$593.5M is nearly double Spielberg's \$323.6M, yet his rating of 3.69 ranks second-lowest in the group. Scorsese, by contrast, earns the highest audience rating (3.76) on a comparatively modest \$115.7M — pointing to two distinct success models: mass-market spectacle versus critically resonant storytelling.

David Fincher is the standout value director. With a rating of 3.72 and \$213.9M in mean revenue, Fincher occupies a rare middle ground — commercially strong *and* audience-approved — making him the most balanced performer in the group.

Rating variance among top directors is surprisingly narrow. All five directors fall within a 0.15-point band (3.61–3.76), indicating that once directors clear the top-quartile threshold, revenue — not rating — becomes the true differentiator.

Walt Disney Pictures leads studios by a wide margin. At \$356.4M mean revenue, Disney outpaces second-ranked Summit Entertainment (\$214.4M) by over 65%, reflecting the unmatched commercial power of its franchise-driven model.

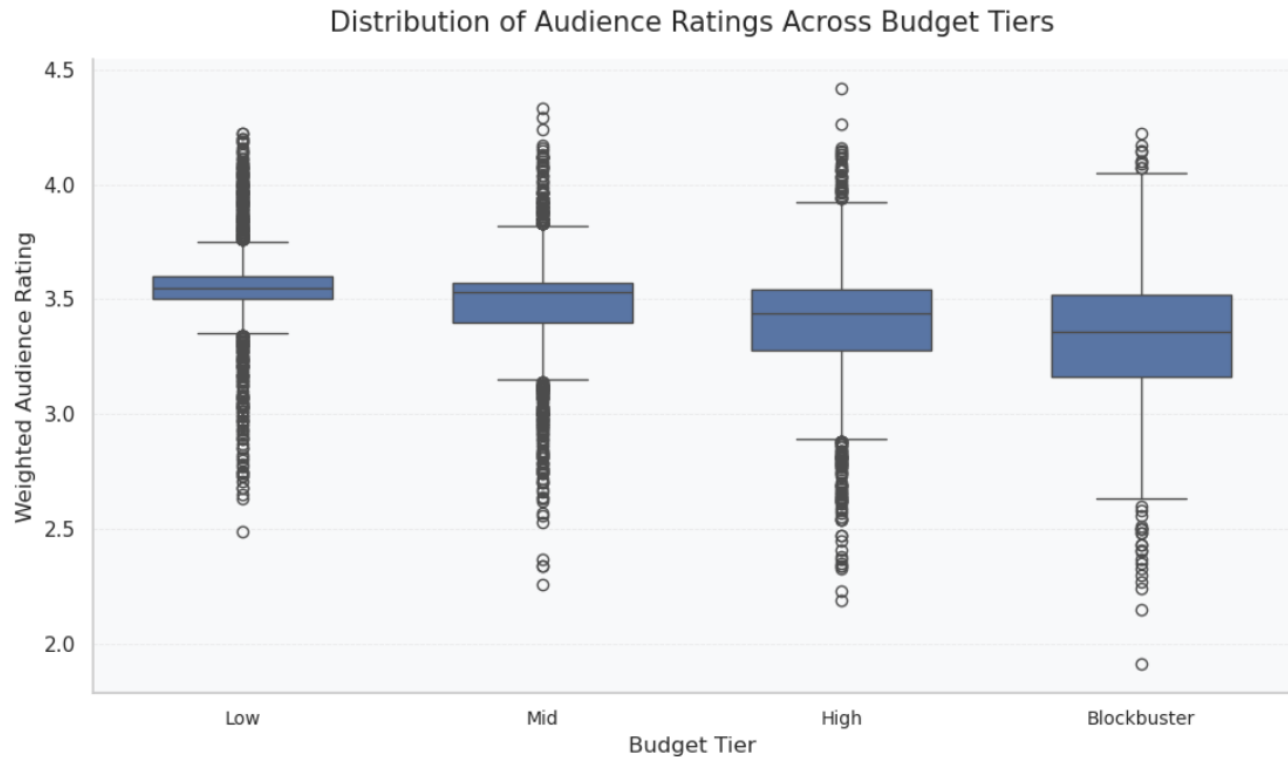
DreamWorks SKG offers the best rating-to-revenue balance among studios. At \$202.3M and a rating of 3.42 — the highest in the group — it mirrors the Fincher pattern at the studio level: efficient, quality-consistent, and commercially competitive without relying on franchise scale.

6.4 Audience Rating Distribution Across Budget Tiers

This analysis examines how **audience reception varies across different movie budget categories**. By comparing the distribution of weighted audience ratings for **low-budget, mid-budget, high-budget, and blockbuster films**, the analysis provides insight into **whether higher production budgets are associated with stronger audience appreciation**.

The following boxplot illustrates the distribution of weighted ratings across the different budget tiers.

Audience Rating Distribution across Budget Tiers



Insights from the Boxplot

The data suggests an **inverse relationship between production budget and audience ratings**, with low-budget films consistently **outperforming their higher-budget counterparts** in median weighted ratings. Budget size appears to increase variability in reception more than it guarantees quality, **challenging the assumption that larger investments lead to better audience outcomes**.

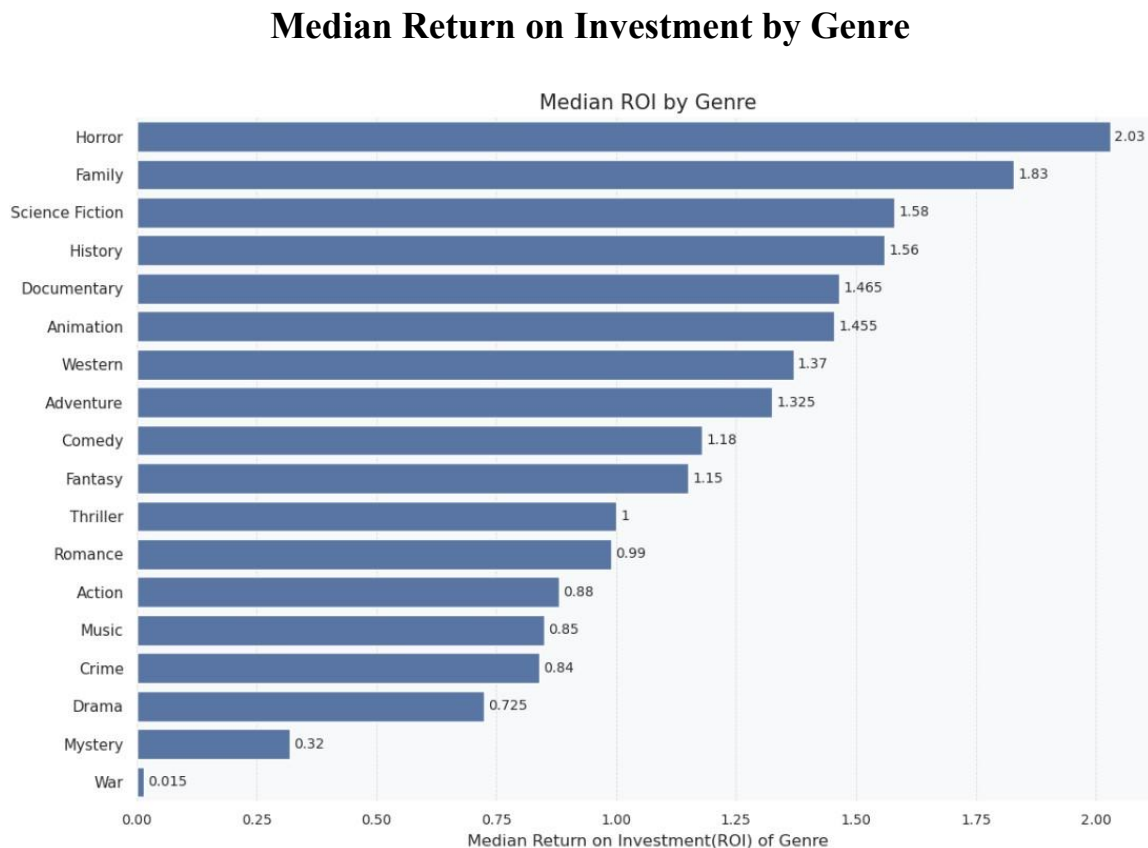
The results imply that audience appreciation may be influenced by several additional factors beyond budget alone, such as **storytelling quality, direction, cast performance, genre, and production studio**.

6.5 Genre Efficiency: Which Genres Deliver the Best ROI?

Box office revenue alone is an **incomplete measure of financial success**. A film that earns \$200M on a \$180M budget is far less efficient than one that earns \$50M on a \$10M budget. This section shifts focus from raw revenue to **median Return on Investment (ROI)** across genres, revealing which genres consistently generate the most financial value relative to their production cost.

Median ROI was chosen over mean to prevent a **small number of breakout blockbusters** from **distorting genre-level conclusions**. To ensure **statistical reliability**, only genres with a **minimum of 25 films** in the dataset were included in the comparison.

The following Bar chart shows the genres by their median ROI:



Insights from ROI analysis

Horror and Family lead on ROI efficiency, posting medians of 2.03 and 1.83 respectively — the only genres to exceed a 1.8x return. Horror's structurally low budgets make it the single most capital-efficient genre in the dataset.

High-revenue genres are poor ROI performers. Action (0.88), Crime (0.84), and Drama (0.725) all fall below breakeven at the median, revealing a fundamental trade-off: the genres that dominate box office headlines consume budgets too large to recover efficiently.

War and Mystery carry the highest financial risk, with median ROIs of just 0.015 and 0.32 — meaning the typical film in these genres fails to recover even half its investment.

The bottom line: genre is a meaningful financial lever. Leaner genres — Horror (2.03), Documentary (1.465), History (1.56) — consistently outperform big-budget counterparts on ROI, making them the stronger choice for capital-efficient filmmaking.

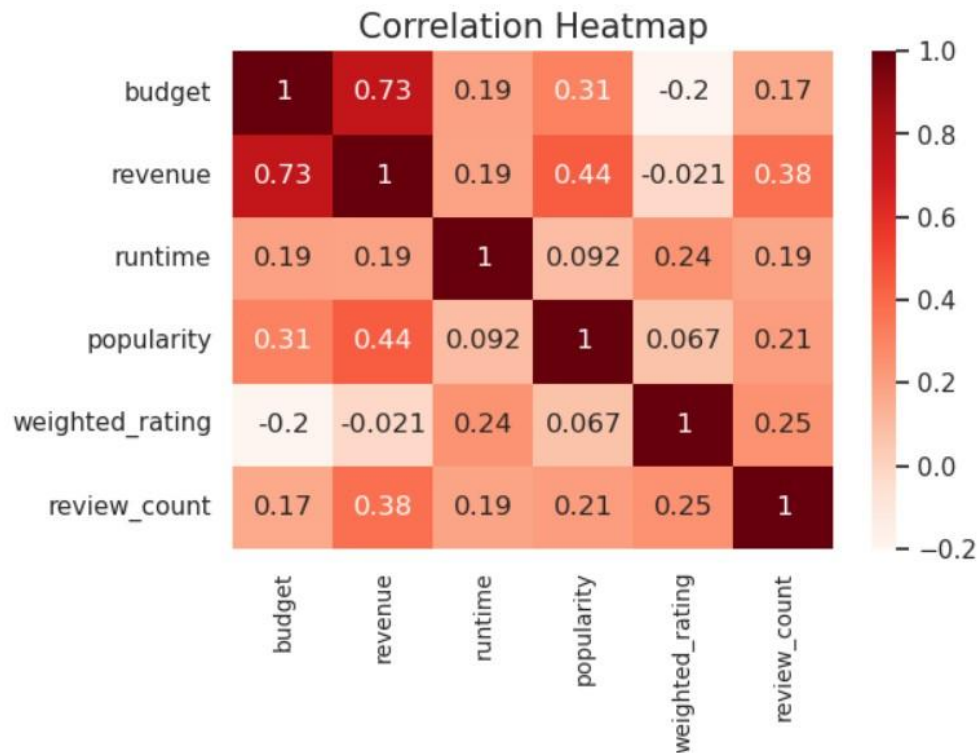
6.6 Variable Correlation Analysis

Before modeling, it is essential to understand how the key variables relate to one another. This section presents a **Pearson correlation matrix** across six variables central to the project's research question — **budget, revenue, popularity, weighted rating, review count, and runtime** — to **quantify the direction and strength of pairwise relationships**.

Beyond informing feature selection for **the regression model**, this analysis helps distinguish which **factors are genuinely associated with financial success (revenue) versus audience reception (weighted rating)**, and flags **potential multicollinearity concerns** where strongly correlated predictors may overlap in explanatory power.

The below **Correlation Heatmap** shows the relationships between Variables:

Correlation Heatmap of Variables



Insights from Correlation Heatmap

Budget and revenue share the strongest relationship ($r = 0.73$), confirming that higher-spending films tend to earn more — but this is a commercial, not creative, driver. It also signals a multicollinearity risk if both are included as predictors in the same model.

Popularity is a meaningful revenue predictor ($r = 0.44$), more so than budget alone suggests, indicating that audience buzz and reach contribute independently to box office performance.

Weighted rating is largely decoupled from financial variables. Its near-zero correlation with revenue ($r = -0.021$) and weak negative correlation with budget ($r = -0.2$) reinforces the earlier finding that financial scale does not drive — and may slightly inhibit — audience approval.

Runtime has negligible correlations across all variables (all below 0.25), suggesting it has little practical influence on either commercial performance or audience reception.

The bottom line: revenue and weighted rating respond to largely different sets of drivers — making them meaningfully distinct target variables that warrant separate modeling treatment.

7.0 Hypothesis Testing

7.1 Financial Success Tests

7.1.1 Budget tier vs Revenue

To statistically evaluate whether movie budget levels influence box office performance, a **one-way ANOVA test was conducted** between budget tiers and movie revenue. The objective of this analysis was to determine **whether the differences in average revenue across budget categories are statistically significant**.

Null Hypothesis (H_0): There is no significant difference in mean revenue across different budget tiers.

Alternative Hypothesis (H_1): At least one budget tier has a significantly different mean revenue.

Results of One-Way ANOVA Test

- F-statistic = 727.18
- P-value < 0.001

The ANOVA results indicate a **statistically significant difference in mean box office revenue across the different budget tiers**. The extremely small p-value suggests that the observed differences are highly unlikely to have occurred by random chance alone. Therefore, **the null hypothesis is rejected**, indicating that movie budget tier has a significant effect on box office revenue.

These findings support the conclusion that higher-budget movies tend to generate substantially different financial outcomes compared to lower-budget films.

Post-Hoc Analysis: Tukey HSD Test

Since the one-way ANOVA test indicated statistically significant differences in mean revenue across budget tiers, a **Tukey HSD post-hoc test was conducted** to identify which specific budget tier pairs differed significantly from one another.

The results of the Tukey HSD test **showed that all pairwise comparisons between budget tiers were statistically significant at the 5% significance level ($p < 0.05$)**. In particular, blockbuster movies generated significantly higher average revenues compared to high, mid, and low-budget

films. Similarly, high-budget movies significantly outperformed both mid-budget and low-budget movies in terms of average box office revenue.

These findings suggest a **strong positive relationship between movie budget tier and financial performance**, with higher-budget productions consistently generating greater box office revenues on average.

7.1.2 Release Season vs Revenue

This analysis investigates whether **movie release timing significantly influences box office revenue**. A **one-way ANOVA test** was conducted to compare the mean revenues of movies released during the Summer, Year-End, and Other release periods, and to **determine whether the observed differences are statistically significant**.

Null Hypothesis (H_0): There is no significant difference in mean box office revenue across the different release seasons.

Alternative Hypothesis (H_1): At least one release season has a significantly different mean box office revenue.

Results of One-Way ANOVA Test

- F-statistic = 37.63
- P-value < 0.001

The ANOVA results indicate that average **movie revenues differ significantly** across release seasons. Since the **p-value is substantially lower than the 5% significance level, the null hypothesis is rejected**. This suggests that release timing plays a statistically significant role in influencing box office performance.

In particular, **movies released during the Summer and Year-End periods appear to achieve stronger financial performance** compared to movies released during other parts of the year.

Post-Hoc Analysis: Tukey HSD Test

To further investigate which release seasons differed significantly from one another, a **Tukey HSD post-hoc test** was performed following the ANOVA analysis.

Insights from Post-Hoc analysis

- **Summer releases** generated significantly **higher average revenues** than movies released during the **Other season**.
- **Year-End releases** also significantly **outperformed Other season releases** in terms of **average revenue**.
- However, **the difference in average revenue between Summer and Year-End releases was not statistically significant ($p = 0.424$)**.

These findings suggest that **movies released during peak theatrical windows such as Summer and Year-End tend to achieve higher box office revenues**, while the **financial performance between these two peak seasons remains statistically similar**.

7.2 Audience Reception Tests

7.2.1 Studio Type vs Weighted Ratings

The **audience reception of a movie may be influenced by the reputation and scale of its production studio**. Larger and more established studios often possess greater financial resources, stronger marketing capabilities, and wider audience reach, which **may affect how their movies are perceived by viewers**.

To investigate this relationship, an **exploratory analysis** was first conducted to compare the average weighted audience ratings of movies produced by Major Studios and Independent Studios. Subsequently, **Welch's independent two-sample t-test** was performed to **statistically evaluate whether the observed difference in ratings was significant**.

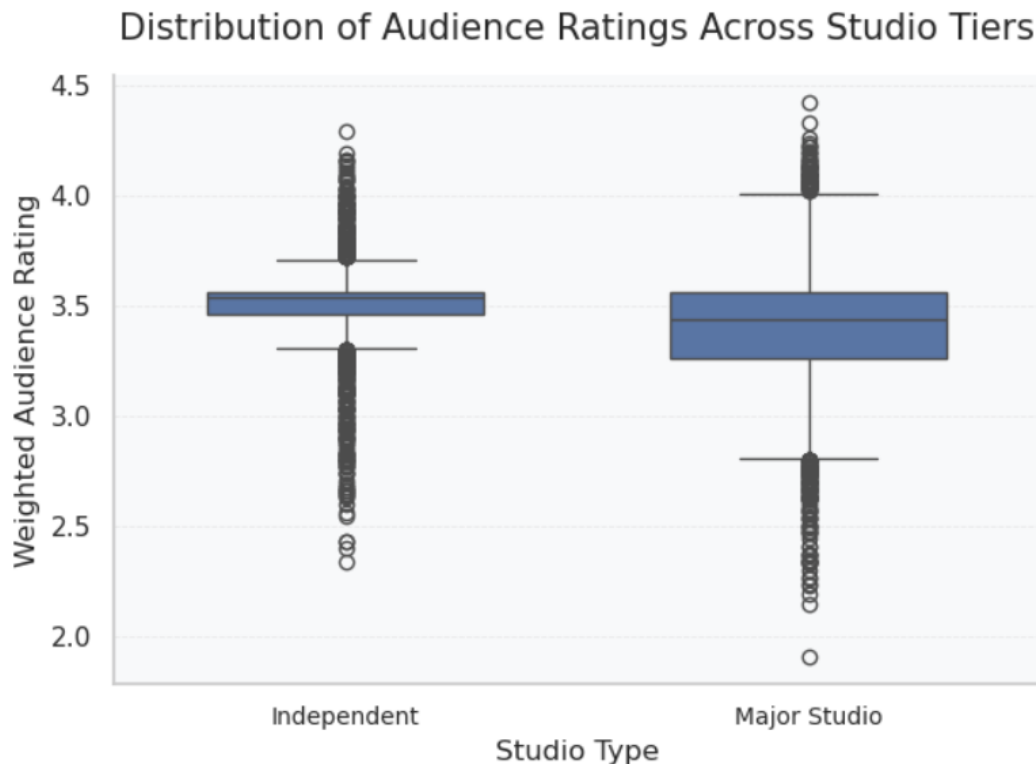
For the purpose of this analysis, studios producing at least five movies in the dataset were categorized as **Major Studios**, while the remaining studios were classified as **Independent Studios**. Additionally, only movies with a minimum review count of 500 were included to ensure the **reliability of audience ratings**.

Analysis and Results

Studio Type	Mean Weighted Rating
Independent Studios	3.49
Major Studios	3.39

The exploratory analysis indicates that **movies produced by Independent Studios achieved slightly higher average weighted ratings compared to movies produced by Major Studios.** The boxplot below visually illustrates the difference in rating distributions between the two studio categories.

Boxplot of Weighted Rating by Studio Type



Independent Two-Sample t-test (Welch's t-test)

Null Hypothesis (H_0): There is **no statistically significant difference** in the mean weighted audience ratings between movies produced by Major Studios and Independent Studios.

Alternate Hypothesis (H_1): There is a **statistically significant difference** in the mean weighted audience ratings between movies produced by Major Studios and Independent Studios.

Results of Welch's t-test

- T-statistic = -7.05
- P-value < 0.001

The results of Welch's t-test indicate that the difference in weighted audience ratings between **Major and Independent Studios** is **statistically significant** at the **5% significance level**. Since the p-value is substantially lower than 0.05, **the null hypothesis of equal mean ratings between the two studio categories is rejected**.

Interestingly, movies **produced by Independent Studios** demonstrate **marginally higher audience ratings** on average than those **produced by Major Studios**. This suggests that while major production houses may achieve stronger commercial performance, **independent studios** may produce films that **receive comparatively greater audience appreciation**.

7.2.2 Genre vs Weighted Ratings

Audience perception of movies may vary across genres due to **differences in viewer preferences and storytelling styles**. To examine whether audience ratings differ significantly across movie genres, **a one-way ANOVA test was conducted using weighted audience ratings**.

To improve the reliability of the analysis, **only movies with at least 500 reviews were included**. Additionally, **genres containing fewer than 25 movies were excluded to avoid unstable group estimates and improve statistical robustness**.

Exploratory Analysis

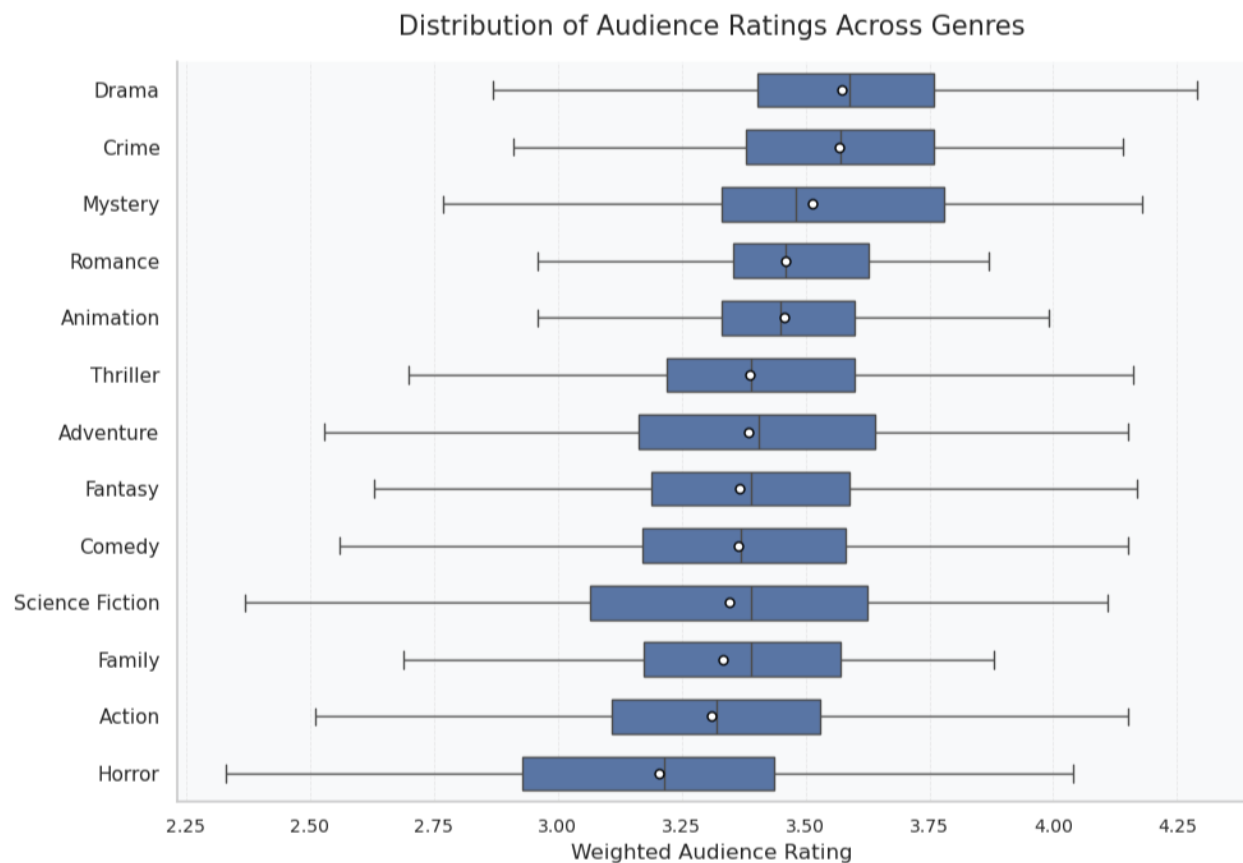
The table below summarizes the average weighted ratings and movie counts for the genres included in the analysis.

Genres	Number of Movies	Average Weighted Rating
Drama	718	3.57
Crime	170	3.57
Mystery	41	3.51
Romance	62	3.46
Animation	97	3.46
Thriller	113	3.39
Adventure	306	3.38
Fantasy	105	3.37
Comedy	625	3.36
Science Fiction	87	3.35
Family	35	3.33
Action	596	3.31
Horror	202	3.20

The exploratory analysis suggests that genres such as **Drama**, **Crime**, and **Mystery** tend to receive **higher audience ratings on average**, while **Horror** and **Action** exhibit **comparatively lower ratings**.

The following boxplots illustrate the **distribution of weighted audience ratings across genres**. **Mean ratings** are represented using **white markers** to highlight **differences in group averages**.

Boxplot of Weighted Rating across Genres



The boxplots visually indicate that higher-rated genres such as **Drama** and **Crime** generally have **rating distributions centered at higher values**, whereas **Horror** displays a **comparatively lower median and mean rating distribution**.

One-Way ANOVA test

Null Hypothesis (H_0): There is no significant difference in mean weighted rating across movie genres.

Alternative Hypothesis (H_1): At least one genre has a significantly different mean weighted rating.

Results of One-Way ANOVA Test

- F-statistic = 33.4709
- P-value < 0.001

The ANOVA results indicate a **statistically significant difference** in average weighted audience ratings across movie genres. Since the p-value is substantially **lower than the 5% significance level**, the **null hypothesis is rejected**. This suggests that genre has a significant influence on audience reception.

Post-Hoc Analysis: Tukey HSD Test

Following the significant ANOVA results, a **Tukey HSD post-hoc test** was conducted to identify which genre pairs differed significantly in average weighted audience ratings.

The post-hoc analysis revealed that genres such as **Drama** and **Crime** generally received significantly **higher audience ratings** compared to genres like **Horror**, **Fantasy**, **Science Fiction**, and **Thriller**. In particular, Horror movies consistently showed lower average audience ratings relative to several other genres.

However, many genre pairs did not exhibit statistically significant differences, suggesting that audience reception overlaps considerably across several movie genres.

These findings indicate that genre has a **measurable influence on audience reception**, although the **magnitude of differences varies across genre combinations**.

8.0 Regression Analysis

8.1 Simple Linear Regression

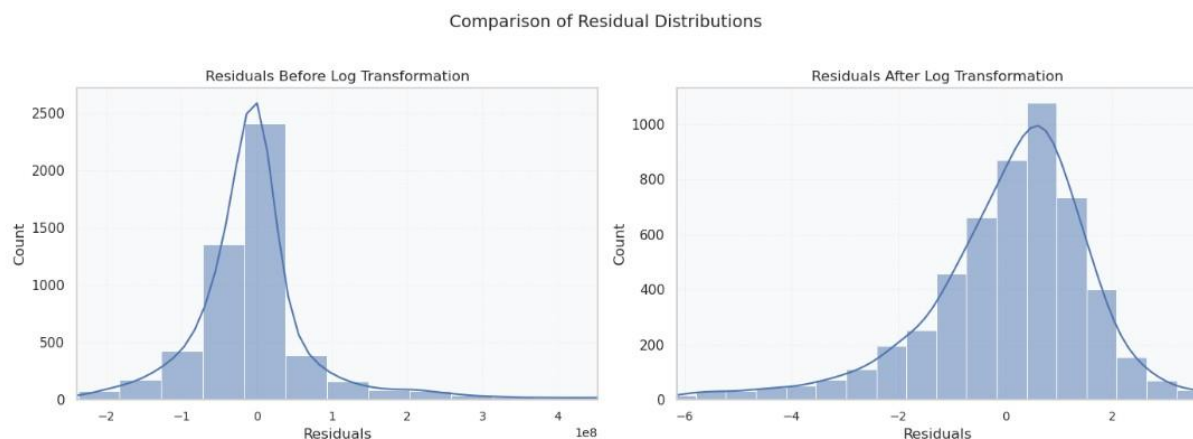
To examine the **relationship between movie production budget and box office revenue**, a **Simple Linear Regression model** was developed using the **Ordinary Least Squares (OLS)** method. The objective of this analysis was to **quantify the extent to which production budget influences a movie's financial performance**.

Initial exploratory analysis revealed that both **budget and revenue distributions were heavily right-skewed** due to the presence of blockbuster movies and extreme revenue outliers. Such skewness **can violate key OLS regression assumptions**, particularly the **normality of residuals** and **homoscedasticity**. To address this issue, **log transformed variables** were used in model development instead of raw variables.

The transformed log-log model produced substantially **improved regression diagnostics**, resulting in a more **stable and statistically reliable model**.

The histograms below compare the residual distributions **before** and **after** transformation and illustrate the improvement in **residual symmetry** and **approximate normality** achieved through logarithmic transformation.

Comparison of Residual Distribution



The residual distribution after transformation appears significantly more symmetric and closer to a normal distribution compared to the original model, indicating improved compliance with OLS regression assumptions.

Model Equation:

$$\text{Log (Revenue)} = 2.3997 + 0.882 \text{ Log (Budget)}$$

Variable	Coefficient	p-value	95% CI
log_budget	0.882	<0.001	[0.858, 0.906]

Model Interpretation:

- The **coefficient of log budget (0.882)** indicates that a **1% increase in movie budget** is associated with an **approximate 0.88% increase in movie revenue on average**.
- The model achieved an **R² value of 0.493**, indicating that **nearly 49.3% of the variation in movie revenue is explained by production budget alone**.
- We are **95% confident that a 1% increase in budget** is associated with approximately a **0.86% to 0.91% increase in revenue**.
- The **regression results** show that movie budget is a **highly significant predictor of revenue (p < 0.001)**, indicating that **production budget** is one of the **major factors influencing a movie's financial performance**.

Model Diagnostics

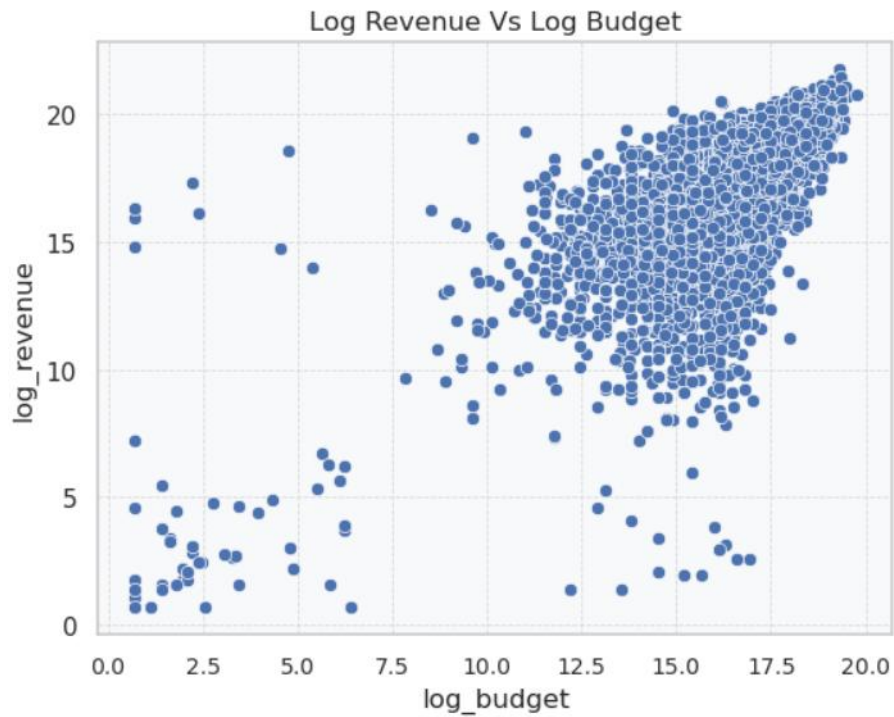
This section analyzes whether the model **satisfies the four assumptions of Simple Linear Regression** given below -

- Linearity
- Independent Observations
- Normality
- Homoscedasticity

Linearity Assumption

The **linearity assumption** states that the **dependent and independent variables of the simple linear regression model are linearly related to each other**. This is verified using the scatter plot between log_revenue variable and log_budget variable as given below-

Scatter Plot of log revenue Vs log budget

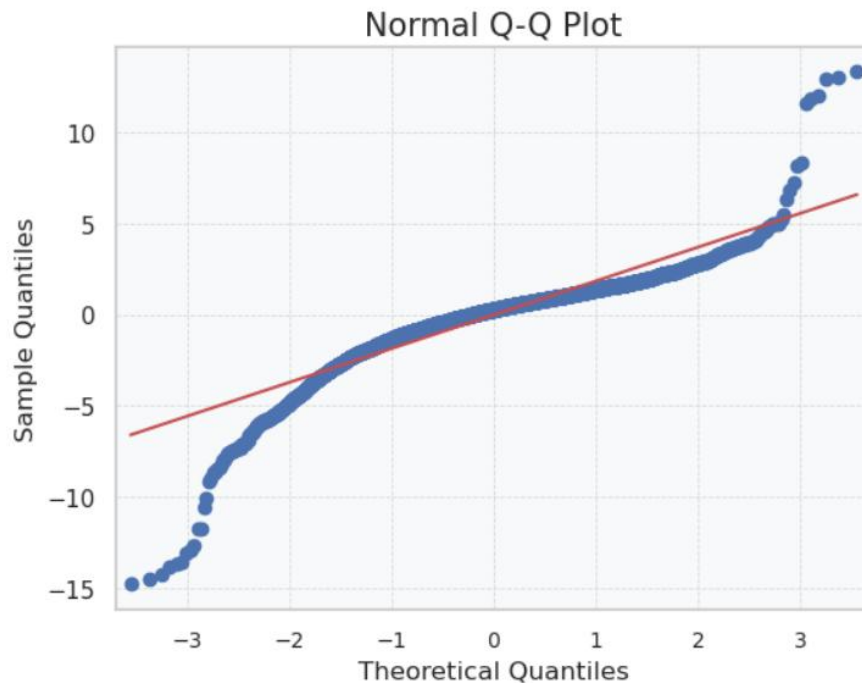


- There is a **clear linear relationship between log_budget and log_revenue** as seen from the scatterplot.
- The **Pearson correlation coefficient** was also checked between these variables which was found to be **0.70** further confirming the linear relationship.
- Thus, it can be said that **linearity assumption is met**.

Normality Assumption

The normality assumption states that the errors are normally distributed. This is verified using the Q-Q plot of residuals as given below-

Q-Q Plot of Residuals



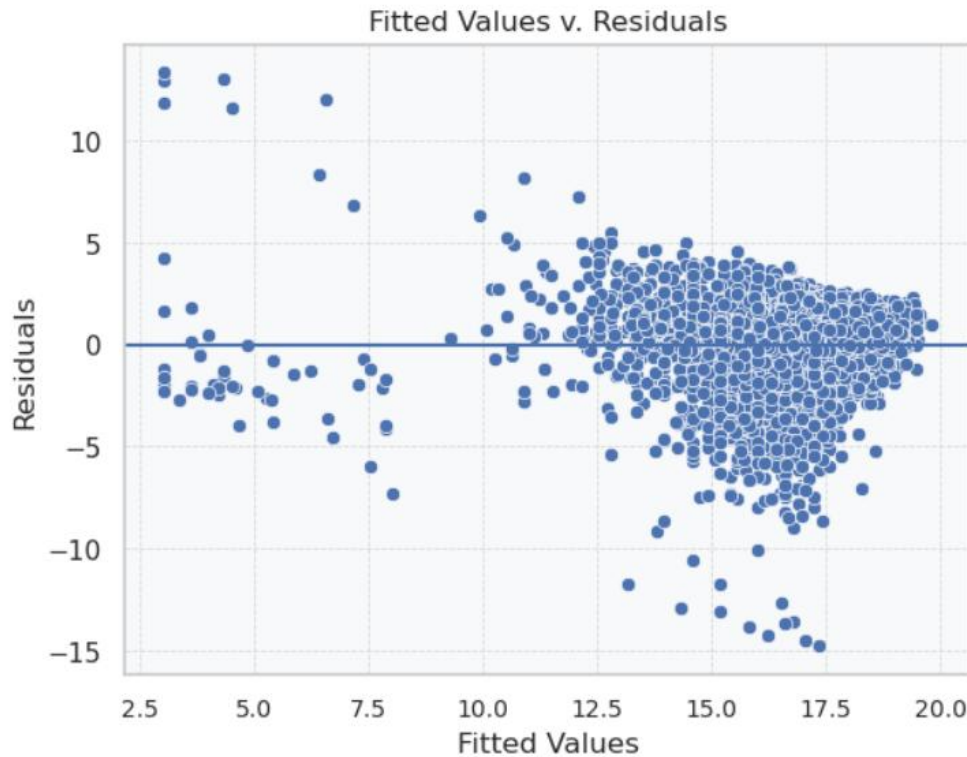
- The Q-Q plot showed that most residuals aligned reasonably well with the reference line in the central region, indicating approximate normality for the majority of observations.
- However, noticeable deviations were still present in the tails, suggesting the presence of some extreme observations and remaining non-normality.
- Such tail deviations are common in movie revenue datasets due to blockbuster outliers and the highly variable nature of box office performance.
- Thus, the model can be said to comply to the Normality Assumption.

Homoscedasticity Assumption

The **homoscedasticity (constant variance) assumption** is that the residuals have a constant variance for all values of X .

This section checks that **this assumption is not violated by creating a scatterplot with the fitted values and residuals**. Add a line at $y = 0$ to visualize the variance of residuals above and below $y = 0$.

Scatterplot of Residuals vs Fitted Values



The **variance of the residuals** is consistent across all **X** values. Thus, the **assumption of homoscedasticity** is met.

Independent Observations Assumption

The **independent observation assumption** states that each observation in the dataset is independent. As a movie's revenue is independent from other movies' revenue, the independence assumption is not violated.

8.2 Multiple Linear Regression

To further investigate the factors influencing movie financial performance, a **Multiple Linear Regression** model was developed using **log-transformed revenue** as the dependent variable and the following predictors:

- Log-transformed production budget (**log_budget**)
- Primary movie genre (**primary_genre**)

- Studio tier (studio_tier)

Genres with fewer than 100 movies were excluded from the analysis to improve statistical reliability and reduce instability caused by sparsely represented categories. Drama was selected as the baseline reference genre for coefficient interpretation.

Model Performance

The Multiple Linear Regression model achieved:

- $R^2 = 0.512$
- Adjusted $R^2 = 0.511$

This indicates that approximately **51.2% of the variation in movie revenue** is explained collectively by budget, genre, and studio tier.

Key Regression Results

Variable	Coefficient	p-value	95% Confidence Interval	Interpretation
log_budget	0.820	<0.001	[0.793, 0.847]	Strong positive relationship with revenue
studio_tier (Major Studio)	0.776	<0.001	[0.667, 0.884]	Major studio movies tend to earn higher revenue
Adventure	0.506	<0.001	[0.309, 0.703]	Higher expected revenue than Drama movies
Animation	0.640	<0.001	[0.338, 0.943]	Strong positive revenue association relative to Drama
Horror	0.582	<0.001	[0.368, 0.796]	Higher expected revenue than Drama movies
Science Fiction	0.450	0.011	[0.102, 0.799]	Positive association with revenue relative to Drama

Genre coefficients are interpreted relative to **Drama movies**, which were used as the **baseline category**.

Model Equation

The estimated regression model can be expressed as:

$$\text{Log (Revenue)} = 2.74 + 0.820 \text{ Log (Budget)} + \text{Genre Effects} + 0.776 \text{ (Studio Tier)} + \epsilon$$

Model Interpretation and Insights

- **Production budget** remained the **strongest predictor** of movie revenue even after controlling for genre and studio tier.
- The coefficient for **log_budget (0.820)** indicates that a **1% increase in production budget is associated with an approximate 0.82% increase in movie revenue**, holding genre and studio tier constant.
- The coefficient for **studio tier (0.776)** implies that holding budget and genre constant, movies produced by **major studios** are associated with approximately **117% higher expected box office revenue** compared to **movies produced by independent studios**.
- The coefficient for **Adventure (0.506)** suggests that holding budget and studio tier constant, Adventure movies are associated with approximately **66% higher expected box office revenue** compared to **Drama movies**.
- Most predictor variables were **statistically significant (p < 0.05)**, suggesting that budget, studio tier, and several movie genres have **meaningful associations with box office revenue**.
- Several genres demonstrated **significantly higher expected revenues** relative to Drama movies. In particular, **Animation, Horror, Adventure, and Science Fiction** movies showed **strong positive** associations with **box office revenue**.

Comparison with Simple Linear Regression

Model	Predictors	Adj R ²
Simple Linear Regression	Log Budget	0.493
Multiple Linear Regression	Log Budget + Genre + Studio Tier	0.512

The inclusion of genre and studio tier improved the model's explanatory power from **49.3% to 51.2%**, indicating that these variables contribute additional predictive value beyond production budget alone. However, the relatively modest improvement also suggests that production budget remains the dominant driver of financial performance in the dataset.

Multi-Collinearity Assumption

The **no multicollinearity assumption** states that no two independent variables (X1 and X2) can be highly correlated with each other.

To check if the multiple regression model complies with this assumption, the **variance inflation factor** method was used to detect multicollinearity.

General VIF Interpretation Guidelines

VIF	Interpretation
1	No multicollinearity
1 – 5	Low to moderate multicollinearity
5 – 10	High multicollinearity
> 10	Serious multicollinearity concern

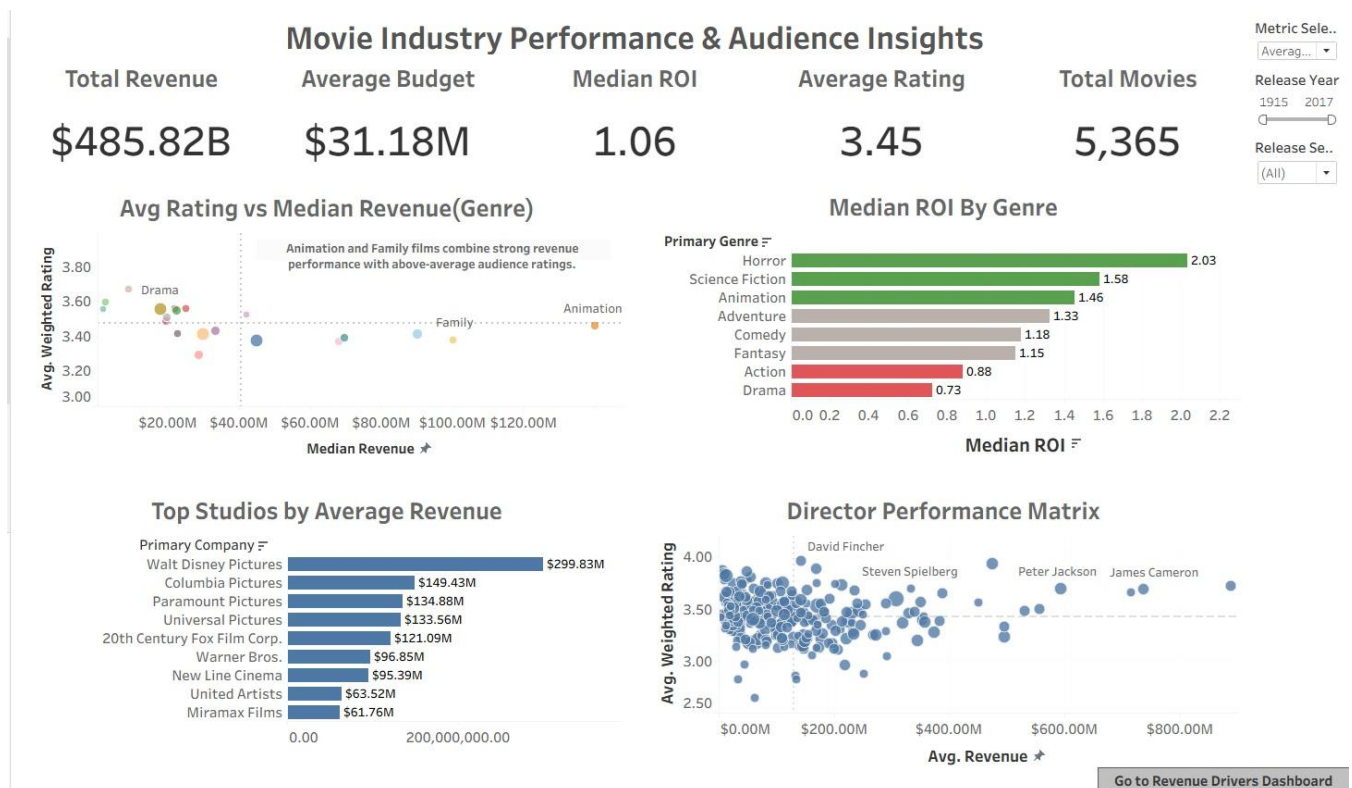
Analysis Results

All predictor variables showed VIF values close to 1 and well below the commonly accepted threshold of 5, indicating that **multicollinearity was not a significant concern in the regression model**.

9.0 Tableau Dashboards

The following dashboards visually corroborate the Analysis done in Python by displaying the most important factors in determining financial success and Audience reception of a movie.

Industry Performance and Audience Insights Dashboard:



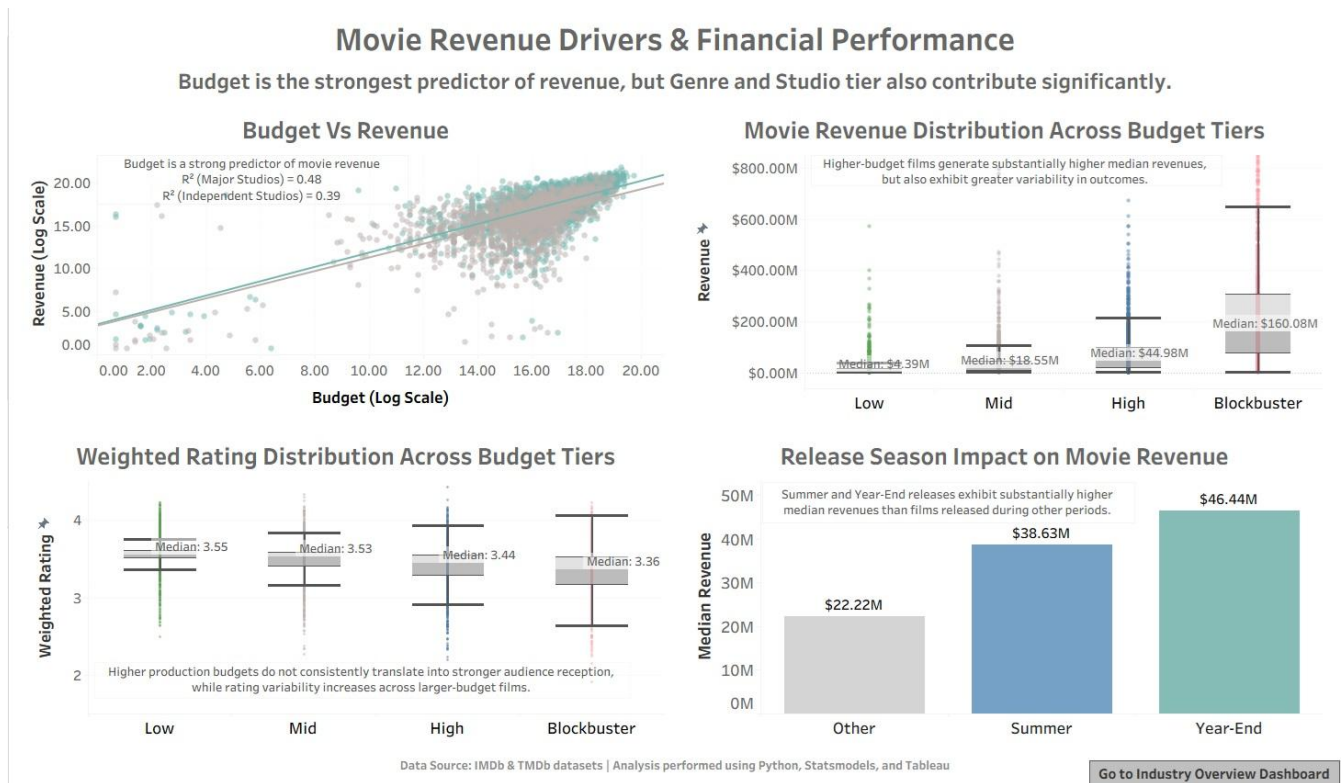
This dashboard provides a high-level overview of the movie industry landscape and consolidates the core findings from the EDA into a single, interactive view.

Key Insights

- KPI Summary Strip** — displays aggregate metrics (Total Revenue: \$485.82B, Average Budget: \$31.18M, Median ROI: 1.06, Average Rating: 3.45, Total Movies: 5,365) providing immediate context on the scale and financial norms of the dataset.

- **Avg Rating vs Median Revenue by Genre** — a scatter plot revealing the **divergence** between **critical appreciation** and **commercial performance**; **Animation** and **Family** occupy the **high-revenue quadrant** while **Drama** clusters **near the high-rating axis**, reinforcing the **genre trade-off** finding.
- **Median ROI by Genre** — a bar chart confirming **Horror (2.03)** and **Science Fiction (1.58)** as the **most capital-efficient genres**, while **Action (0.88)** and **Drama (0.73)** fall below breakeven at the median.
- **Top Studios by Average Revenue** — a ranked bar chart showing **Walt Disney Pictures (\$299.83M)** leading by a substantial margin, followed by **Columbia Pictures** and **Paramount Pictures**.
- **Director Performance Matrix** — a scatter plot mapping directors on average revenue versus average weighted rating, enabling simultaneous evaluation of commercial and creative performance. **David Fincher's** position in the **upper-middle quadrant** visually captures his **dual strength**, while **Peter Jackson** and **James Cameron** anchor the high-revenue, moderate-rating cluster.
- **Interactive Controls** — **Release year slider** and **Release season filter** allow stakeholders to isolate trends across specific time periods and theatrical windows.

Movie Revenue Drivers and Financial Performance Dashboard:



This dashboard focuses specifically on the **quantitative drivers of box office revenue** and directly supports the **regression** and **hypothesis testing findings**.

Key Insights

- **Budget vs Revenue Scatter Plot (Log Scale)** — overlays separate trend lines for Major Studios ($R^2 = 0.48$) and Independent Studios ($R^2 = 0.39$), illustrating that budget is a stronger revenue predictor within major studio productions and that studio tier modulates the budget–revenue relationship.
- **Movie Revenue Distribution Across Budget Tiers** — box plots showing median revenues escalating from **\$4.39M (Low)** through **\$18.55M (Mid)** and **\$44.98M (High)** to **\$160M (Blockbuster)**, with visibly increasing spread at higher tiers — corroborating the ANOVA finding while highlighting growing outcome variability.
- **Weighted Rating Distribution Across Budget Tiers** — box plots with near-flat medians (3.55 → 3.53 → 3.44 → 3.36) across budget tiers, providing a clear visual counterpoint: budget scales revenue but not audience approval.

- **Release Season Impact on Movie Revenue** — a bar chart comparing median revenues across seasons (Other: \$22.22M, Summer: \$38.63M, Year-End: \$46.44M), confirming the statistical significance of peak-window releases and making the magnitude of the timing premium immediately interpretable for non-technical audiences.

10.0 Summary of Insights

➤ **Budget is the dominant financial lever, but not a creative one.**

A 1% increase in production budget is associated with a **0.82% increase in revenue ($R^2 = 0.51$)**, making budget the single strongest predictor of box office success. However, audience ratings decline slightly as budget tiers increase, indicating that higher spending does not guarantee stronger audience reception.

➤ **Genre determines the trade-off between revenue and ROI.**

Animation, Family, and Adventure generate the highest median revenues (\$147M–\$246M), while **Horror (ROI: 2.03)** and **Family (ROI: 1.83)** deliver the strongest return on investment. In contrast, **Action (ROI: 0.88)** and **Drama (ROI: 0.73)** underperform on capital efficiency despite substantial budgets.

➤ **Studio tier is a statistically significant revenue multiplier.**

After controlling budget and genre, major studio releases generate approximately **117% higher expected revenue** than independent films, highlighting the impact of distribution reach, marketing scale, and franchise ecosystems.

➤ **Release timing influences revenue independently of budget.**

Movies released during **Summer and Year-End windows** achieve substantially higher median revenues (\$38.6M–\$46.4M) than off-peak releases (\$22.2M), demonstrating that timing contributes materially to box office performance.

➤ **Audience reception follows a different formula than commercial success.**

Audience ratings show virtually **no relationship with revenue ($r = -0.02$)** and a weak negative relationship with budget ($r = -0.20$). Independent films and genres such as **Documentary, Western, War, Drama, and Crime** consistently receive stronger audience ratings despite generating lower box office returns.

11.0 Business Recommendations

- **Maximize Box Office Revenue:** Focus investment on **Animation, Family, and Adventure** titles, leverage **major studio distribution**, and target **Summer or Year-End** release windows. This combination is **most strongly associated** with **higher revenue outcomes**.
- **Maximize ROI:** Prioritize **Horror** and **Documentary** projects, which deliver the **strongest capital efficiency**. **Horror** offers **attractive returns** due to its low production costs and consistent audience demand.
- **Maximize Audience Satisfaction:** Invest in **Drama, Crime, and Mystery** films and **emphasize strong creative leadership**. Audience ratings are driven **more by genre and storytelling quality** than by **budget size or studio scale**, with **independent productions** often **outperforming major studios** on audience reception.
- **Optimize Release Strategy:** Since **Summer and Year-End** releases generate **comparable revenues**, select release windows based on market competition, marketing considerations, and production readiness rather than season alone.

12.0 Conclusion

This project analyzed over **42,000 movies** using data cleaning, feature engineering, exploratory analysis, hypothesis testing, and regression modeling to identify the key drivers of **box office performance** and **audience reception**.

The findings show that these outcomes are driven by fundamentally different factors. **Revenue** is primarily influenced by **budget, studio tier, genre, and release timing**, with the model explaining approximately **51% of revenue variation**. In contrast, **audience ratings are largely independent of financial variables** and are more closely associated with genre and creative factors.

Statistical testing confirmed that **budget tier, genre, studio type, and release season** have significant impacts on commercial performance, while the regression models quantified the magnitude of these effects. Most notably, the analysis challenges the assumption that bigger budgets produce better films: **high-budget productions generate higher revenues but not higher audience ratings**, while **Horror outperforms Action on ROI** and **independent films often achieve stronger audience reception than major studio releases**.